

Generalized Graph Signal Reconstruction via the Uncertainty Principle

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Abstract—We introduce a novel uncertainty principle for generalized graph signals that extends classical time-frequency and graph uncertainty principles into a unified framework. By defining joint vertex-time and spectral-frequency spreads, we quantify signal localization across these domains, revealing a trade-off between them. This framework allows us to identify a class of signals with maximal energy concentration in both domains, forming the fundamental atoms for a new joint vertex-time dictionary. This dictionary enhances signal reconstruction under practical constraints, such as intermittent data, commonly encountered in sensor and social networks. Numerical experiments on real-world datasets demonstrate the effectiveness of the proposed approach, showing improved reconstruction accuracy and noise robustness compared to existing methods.

Index Terms—Uncertainty principle, localized representation, joint vertex-time dictionary, generalized graph signal reconstruction

I. INTRODUCTION

The uncertainty principle [1], [2] highlights the trade-off between a signal's time duration and frequency bandwidth, indicating that a signal cannot be precisely localized in both domains simultaneously. Prolate spheroidal wave functions (PSWFs) [3]–[7] optimize time-frequency localization by maximizing energy concentration within a finite time interval while being frequency band-limited, making them effective basis functions for signal representation with minimal information loss. Similarly, an uncertainty principle for graph signals [8] describes a trade-off between localization in the graph and spectral domains. Building on this, [9] introduced a principle that characterizes vertex and spectral spreads, reflecting a signal's energy concentration across specific vertex subsets and spectral components. Unlike regular signals, certain graph signals can be perfectly localized in both domains. An important problem is learning dictionaries with good localization properties, which can enable efficient graph signal representation [10]. In this work, we examine signal localization via energy concentration, though alternative definitions of localization exist [11]–[13].

Traditional graph signal processing (GSP) primarily focuses on information in the graph domain rather than the time domain. The vertex-time GSP theory [14]–[16] addresses this gap by integrating information from both the graph and discrete-time domain. This work has led to joint vertex-time Fourier transforms, filters, harmonic analysis, and redundant

joint vertex-time dictionaries [17], [18] for localized signal representations, such as the short time-vertex Fourier transform (STVFT) and spectral time-vertex wavelet transform (STVWT). Motivated by applications in sensor and social networks, where nodes observe time-varying continuous signals, this theory has been further extended to accommodate vertex signals from an infinite-dimensional, separable Hilbert space [19], termed the generalized graph signal processing (GGSP) framework.

Practical challenges such as limited sensor battery life and environmental factors restrict data availability, complicating signal reconstruction. Efficient data collection under such constraints requires designing basis functions that are maximally energy-concentrated in the joint vertex-time domain. Inspired by uncertainty principles for graph and continuous-time signals, we propose a generalized uncertainty principle to design energy-concentrated basis functions for signal reconstruction.

The contributions of this paper are threefold: a) We introduce a novel uncertainty principle for generalized graph signals, defining joint vertex-time and spectral-frequency spreads to measure a signal's localization in joint vertex-time and spectral-frequency domains. This principle includes existing uncertainty principles for both graph and continuous-time signals as special cases. b) We identify a class of signals maximally concentrated in both joint vertex-time and spectral-frequency domains, use them as a dictionary for signal reconstruction. We derive conditions for perfect localization, along with an optimization framework to enhance reconstruction accuracy. c) We validate our proposed method on real datasets, demonstrating its effectiveness with limited data and its superior noise resilience compared to other methods.

II. BASIC DEFINITIONS

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a simple, connected, undirected graph with N vertices $\mathcal{V} = \{1, 2, \dots, N\}$ and weighted edges $\mathcal{E} = \{a_{ij}\}_{i,j \in \mathcal{V}}$, where $a_{ij} > 0$ if vertices i and j are connected, and $a_{ij} = 0$ otherwise. The adjacency matrix $\mathbf{A} = [a_{ij}]$ is symmetric with zero diagonal entries. The degree matrix $\mathbf{D} = \text{diag}\{d_1, d_2, \dots, d_N\}$, where $d_i = \sum_{j=1}^N a_{ij}$, and the graph Laplacian matrix $\mathbf{L} = \mathbf{D} - \mathbf{A}$ serves as the graph shift operator (GSO). In GSP, a finite-energy graph signal is a function in $L^2(\mathcal{V})$. The Laplacian matrix $\mathbf{L}$ has an eigen-decomposition $\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T = \sum_{i=1}^N \lambda_i \mathbf{u}_i \mathbf{u}_i^T$, where $\mathbf{\Lambda}$ is a diagonal matrix with non-negative real eigenvalues $\{\lambda_i\}_{i=1, \dots, N}$ and $\mathbf{U}$ is the unitary matrix of real-valued orthonormal eigenvectors $\{\mathbf{u}_i\}_{i=1, \dots, N}$. Here, $(\cdot)^T$ denotes

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transpose. The graph Fourier transform (GFT) [20]–[25] of a graph signal $\mathbf{x}$ is given by $\hat{\mathbf{x}} = \mathbf{U}^\top \mathbf{x}$. In GGSP, a generalized graph signal f is a map from the vertex set $\mathcal{V}$ to a separable Hilbert space $\mathcal{H}$. In this work, we consider $\mathcal{H} = L^2(\mathcal{T})$ (with the usual Lebesgue measure), where $\mathcal{T} = \mathbb{R}$ represents the time domain. The signal f is thus a map $\mathcal{V} \times \mathcal{T} \rightarrow \mathbb{R}$, defined by $(v, t) \mapsto f(v, t)$, where $\mathcal{V}$ and $\mathcal{T}$ represent the *vertex* and *time* domains, respectively. The space of generalized graph signals is denoted by $L^2(\mathcal{V} \times \mathcal{T})$. For $f \in L^2(\mathcal{V} \times \mathcal{T})$, the joint Fourier transform (JFT) is defined as $\mathcal{F}_f(\lambda_k, \omega) = \langle f, \mathbf{u}_k \otimes e^{j\omega t} \rangle$ [19], where λ_k and $\mathbf{u}_k$ are the k -th eigenvalue and eigenvector of $\mathbf{L}$, and $\otimes$ is the tensor product. This transform maps f in $L^2(\mathcal{V} \times \mathcal{T})$ having the joint vertex-time domain $\mathcal{V} \times \mathcal{T}$ to a signal in $L^2(\mathcal{W} \times \Omega)$ with the joint spectral-frequency domain $\mathcal{W} \times \Omega$, where $\mathcal{W} = \{\lambda_1, \dots, \lambda_N\}$ and $\Omega = \mathbb{R}$ are the *spectral* and *frequency* domains, respectively. It generalizes both GFT and the classical Fourier transform (FT). Specifically, it reduces to the GFT, which maps a graph signal from $\mathcal{V}$ to $\mathcal{W}$, when f is evaluated at a fixed time instance, and to the FT mapping a temporal signal from $\mathcal{T}$ to Ω when $\mathcal{V}$ is a singleton.

III. UNCERTAINTY PRINCIPLE FOR GENERALIZED GRAPH SIGNALS

A. Joint vertex-time and spectral-frequency spread

We first define the spread in $\mathcal{V} \times \mathcal{T}$ and $\mathcal{W} \times \Omega$ domains, measuring signals' localization in these domains, respectively.

Definition 1. (*Joint vertex-time limiting operator*) Given a subset $\mathcal{S} \subseteq \mathcal{V} \times \mathcal{T}$ and a generalized graph signal $f \in L^2(\mathcal{V} \times \mathcal{T})$, a joint vertex-time limiting operator is defined as $\Pi_{\mathcal{S}} f := f(v, t)$ if $(v, t) \in \mathcal{S}$ and 0 otherwise. $\Pi_{\mathcal{S}}$ is a Hermitian operator, and satisfies $\Pi_{\mathcal{S}}^2 = \Pi_{\mathcal{S}}$.

The joint vertex-time limiting operator $\Pi_{\mathcal{S}}$ generalizes both the vertex-limiting operator $\Pi_{\mathcal{V}'}$, defined by a diagonal matrix $\mathbf{D}_{\mathcal{V}'} = \text{diag}\{\mathbf{1}_{\mathcal{V}'}\}$ for a vertex subset $\mathcal{V}' \subseteq \mathcal{V}$ in the graph domain [9] and the time-limiting operator $\Pi_{\mathcal{T}'}$, restricting a temporal signal $f(t)$ to $\mathcal{T}' = [t_c - \ell/2, t_c + \ell/2] \subseteq \mathcal{T}$, where t_c is the center time point and ℓ is the length of $\mathcal{T}'$ in time domain [3]. Further details are in Remark 1 of [26].

Definition 2. (*Joint spectral-frequency limiting operator*) Given a unitary transform $\mathcal{U} : L^2(\mathcal{V} \times \mathcal{T}) \rightarrow L^2(\mathcal{W} \times \Omega)$ and a subset $\Sigma \subseteq \mathcal{W} \times \Omega$, the joint spectral-frequency limiting operator Π_{Σ} for a generalized graph signal $f \in L^2(\mathcal{V} \times \mathcal{T})$ is defined as $\Pi_{\Sigma} f = \mathcal{U}^{-1} \Sigma \{\mathcal{U} f\}$, where¹ $\Sigma \{\mathcal{U} f\} := (\mathcal{U} f)(w, \omega)$ if $(w, \omega) \in \Sigma$ and 0 otherwise. Π_{Σ} is a Hermitian operator, satisfying $\Pi_{\Sigma}^2 = \Pi_{\Sigma}$.

A typical example of a unitary transform $\mathcal{U}$ mapping from $L^2(\mathcal{V} \times \mathcal{T})$ to $L^2(\mathcal{W} \times \Omega)$ is the JFT. The operator Π_{Σ} generalizes both the spectral-limiting operator $\Pi_{\mathcal{W}'}$ = $\mathbf{U} \Sigma_{\mathcal{W}'} \mathbf{U}^\top$, with $\Sigma_{\mathcal{W}'} = \text{diag}\{\mathbf{1}_{\mathcal{W}'}\}$ and $\mathcal{W}' \subseteq \mathcal{W}$ [9], and the frequency-limiting operator $\Pi_{\Omega'}$ [3], which yields a bandlimited function $\Pi_{\Omega'} f$ within the frequency interval $\Omega' = [\omega_c - W, \omega_c + W]$, where ω_c is the center frequency and W denotes the bandwidth. Further details are provided in Definition 2 of [26].

¹We omit $\mathcal{U}$ in the notation Π_{Σ} to avoid clutter as it will be clear from the context.

The operators $\Pi_{\mathcal{S}}$ and Π_{Σ} are orthogonal projections mapping a generalized graph signal f into the joint vertex-time limiting and joint spectral-frequency limiting spaces, respectively. The image of $\Pi_{\mathcal{S}}$, denoted as $\text{im}(\Pi_{\mathcal{S}})$, contains signals perfectly localized over $\mathcal{S}$, i.e., $\Pi_{\mathcal{S}} f = f$. Similarly, $\text{im}(\Pi_{\Sigma})$ contains signals perfectly localized over Σ , i.e., $\Pi_{\Sigma} f = f$. Let $\bar{\mathcal{S}}$ and $\bar{\Sigma}$ be the complement of $\mathcal{S}$ and Σ , and the projections onto $\bar{\mathcal{S}}$ and $\bar{\Sigma}$ are denoted by $\bar{\Pi}_{\mathcal{S}}$ and $\bar{\Pi}_{\Sigma}$, respectively. Both $\bar{\Pi}_{\mathcal{S}}$ and $\bar{\Pi}_{\Sigma}$ are Hermitian and idempotent. The identities $\Pi_{\mathcal{S}} f + \bar{\Pi}_{\mathcal{S}} f = f$ and $\Pi_{\Sigma} f + \bar{\Pi}_{\Sigma} f = f$ hold.

Definition 3. (*Joint vertex-time spread and joint spectral-frequency spread*) For a generalized graph signal $f \in L^2(\mathcal{V} \times \mathcal{T})$, the joint vertex-time limiting operator $\Pi_{\mathcal{S}}$ and the joint spectral-frequency limiting operator Π_{Σ} project f onto the subsets $\mathcal{S}$ and Σ , respectively. The joint vertex-time spread $\alpha_{\mathcal{S}}^2$ and joint spectral-frequency spread β_{Σ}^2 are defined as $\alpha_{\mathcal{S}}^2 = \frac{\|\Pi_{\mathcal{S}} f\|_2^2}{\|f\|_2^2}$ and $\beta_{\Sigma}^2 = \frac{\|\Pi_{\Sigma} f\|_2^2}{\|f\|_2^2}$. These spreads measure the energy concentration of f within the subsets $\mathcal{S}$ and Σ .

B. Uncertainty principle

An uncertainty principle characterizes the trade-off between energy concentrations $\alpha_{\mathcal{S}}^2$ and β_{Σ}^2 for a vertex-time subset $\mathcal{S}$ and a spectral-frequency subset Σ . This trade-off is presented in the following theorem.

Theorem 1. The feasible region of all possible pairs of $(\alpha_{\mathcal{S}}, \beta_{\Sigma})$ is given by

$$\Theta = \left\{ (\alpha_{\mathcal{S}}, \beta_{\Sigma}) : \begin{aligned} &\cos^{-1} \alpha_{\mathcal{S}} + \cos^{-1} \beta_{\Sigma} \geq \cos^{-1} \sqrt{\lambda_{\max}(\Pi_{\Sigma} \Pi_{\mathcal{S}} \Pi_{\Sigma})}, \\ &\cos^{-1} \sqrt{1 - \alpha_{\mathcal{S}}^2} + \cos^{-1} \beta_{\Sigma} \geq \cos^{-1} \sqrt{\lambda_{\max}(\Pi_{\Sigma} \bar{\Pi}_{\mathcal{S}} \Pi_{\Sigma})}, \\ &\cos^{-1} \alpha_{\mathcal{S}} + \cos^{-1} \sqrt{1 - \beta_{\Sigma}^2} \geq \cos^{-1} \sqrt{\lambda_{\max}(\bar{\Pi}_{\Sigma} \Pi_{\mathcal{S}} \bar{\Pi}_{\Sigma})}, \\ &\cos^{-1} \sqrt{1 - \alpha_{\mathcal{S}}^2} + \cos^{-1} \sqrt{1 - \beta_{\Sigma}^2} \\ &\quad \geq \cos^{-1} \sqrt{\lambda_{\max}(\bar{\Pi}_{\Sigma} \bar{\Pi}_{\mathcal{S}} \bar{\Pi}_{\Sigma})}. \end{aligned} \right\} \quad (1)$$

where $\lambda_{\max}(\cdot)$ denotes the largest eigenvalue of its operator argument.

Proof. See [26]. $\square$

Fig. 1 shows the feasible region Θ with boundary curves from the equality conditions in (1). The upper right corner of Θ represents the pairs $(\alpha_{\mathcal{S}}, \beta_{\Sigma})$ with maximum localization in both vertex-time and spectral-frequency domains. As $\mathcal{S}$ expands for a given Σ , the upper boundary curve progressively approaches a point, indicating projections $\Pi_{\mathcal{S}}$ and Π_{Σ} achieve perfect localization, as detailed in Theorem 2 in Section IV-A.

IV. JOINT VERTEX-TIME DICTIONARY FOR GENERALIZED GRAPH SIGNAL RECONSTRUCTION

Using the uncertainty principle, we construct a dictionary of signals with maximal vertex-time and spectral-frequency concentration, derive localization conditions, and outline the optimization process for reconstruction.

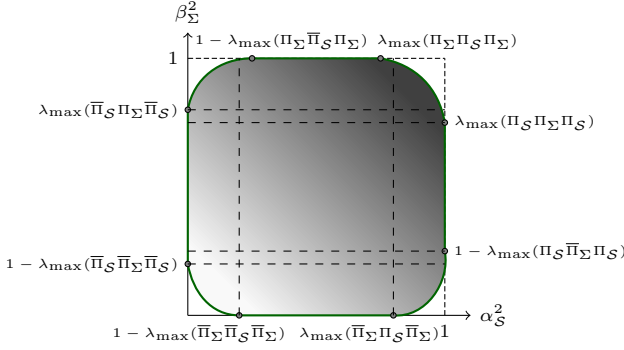


Fig. 1. Feasible region Θ of unit norm generalized graph signals f with $\|\Pi_S f\|_2 = \alpha_S$ and $\|\Pi_\Sigma f\|_2 = \beta_S$

A. Localization properties

We present the perfect localization conditions as follows.

Theorem 2. A nonzero generalized graph signal f can be perfectly localized over both the joint vertex-time subset $\mathcal{S}$ and spectral-frequency subset Σ (i.e., $f \in \text{im}(\Pi_S) \cap \text{im}(\Pi_\Sigma)$) if and only if f is an eigenvector of the operator $\Pi_\Sigma \Pi_S \Pi_\Sigma$ with an eigenvalue of 1, i.e., $\Pi_\Sigma \Pi_S \Pi_\Sigma f = f$.

Proof. See [26]. $\square$

Theorem 2 extends Theorem 2.1 in [9]. To give insights into the perfect localization conditions in Theorem 2, we examine three special cases shown in Remark 2 in [26]. In cases where signals are not perfectly localized in both $\mathcal{S}$ and Σ (e.g., when both $\mathcal{S}$ and Σ are bounded sets), it is crucial to identify signals that are localized in one domain while maximizing their energy concentrations in the other domain. We say that a generalized graph signal is bandlimited if it is localized in the spectral-frequency domain, i.e., it is bandlimited in the usual sense in both the spectral and frequency domains. Our goal is to find an orthonormal set of bandlimited signals, satisfying $\Pi_\Sigma f = f$, that are maximally concentrated in $\mathcal{S}$. This is achieved by solving an optimization problem for a set of signals $\{\xi_i\}_{i=1}^N$, iteratively as follows:

$$\begin{aligned} & \max_{\xi_i} \|\Pi_S \xi_i\| \\ \text{s. t. } & \Pi_\Sigma \xi_i = \xi_i, \|\xi_i\| = 1, \langle \xi_i, \xi_j \rangle = 0, \forall j < i. \end{aligned} \quad (2)$$

The solution to (2) is provided in the following theorem.

Theorem 3. Let $\{\xi_i\}_{i=1,2,\dots}$ be the solution of (2), each maximally concentrated over the vertex-time set $\mathcal{S}$. Then, these signals are the eigenvectors of the operator $\Pi_\Sigma \Pi_S \Pi_\Sigma$, i.e., $\Pi_\Sigma \Pi_S \Pi_\Sigma \xi_i = \lambda_i \xi_i$ where the eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ form a non-increasing sequence. Moreover, these signals are orthogonal over $\mathcal{S}$, i.e., $\langle \xi_i, \Pi_S \xi_j \rangle = \lambda_j \delta_{ij}$, where δ_{ij} is the Kronecker delta.

Proof. See [26]. $\square$

Theorem 3 offers a general framework for constructing orthonormal bandlimited signals $\{\xi_i\}_{i \geq 1}$. As noted in Remark 3 of [26], it includes orthonormal bandlimited vectors $\{\phi_i\}_{i=1,2,\dots,K}$ with $K = \text{rank}(\Pi_{\mathcal{V}'})$, obtained from the eigenvectors of $\Pi_{\mathcal{V}'} \Pi_{\mathcal{V}'} \Pi_{\mathcal{V}'}$ in the graph domain [9], and PSWFs

$\{\psi_j\}_{j \geq 1}$, derived from the eigenfunctions of $\Pi_{\Omega'} \Pi_{\mathcal{T}'} \Pi_{\Omega'}$ in the Hilbert space $L^2(\mathbb{R})$ [3]–[5]. For subsets $\mathcal{S} = \mathcal{V}' \times \mathcal{T}'$ and $\Sigma = \mathcal{W}' \times \Omega'$, the operators Π_S and Π_Σ are given by $\Pi_S = \Pi_{\mathcal{V}'} \otimes \Pi_{\mathcal{T}'}$ and $\Pi_\Sigma = \Pi_{\mathcal{W}'} \otimes \Pi_{\Omega'}$, respectively. The resulting orthonormal bandlimited signals, $\{\xi_{i,j}\}_{i,j \geq 1} = \{\phi_i \otimes \psi_j\}_{i,j \geq 1}$, form a dictionary for reconstructing generalized graph signals.

B. Joint vertex-time dictionary for generalized graph signal reconstruction

An overcomplete dictionary with richer atoms enhances signal reconstruction by enabling sparse, flexible, and robust representations. Motivated by sensor networks with sensors that may have different operational time periods, we use multiple subsets $\mathcal{S}^{(i)}$, $i = 1, 2, \dots, I$, to cover the joint vertex-time domain. For computational feasibility, we consider only subsets of the form $\mathcal{S}^{(i)} = \mathcal{V}'^{(i)} \times \mathcal{T}'^{(i)}$, where $\mathcal{T}'^{(i)} = [t_c^{(i)} - \ell^{(i)}/2, t_c^{(i)} + \ell^{(i)}/2]$. Our objective is to learn a dictionary bandlimited to $\Sigma = \mathcal{W}' \times \Omega'$ and maximally concentrated on a collection of the subsets $\mathcal{S}^{(i)}$. Let $\Phi(\mathcal{V}'^{(i)}) = (\phi_k(\cdot; \mathcal{V}'^{(i)}))_{k \geq 1}$, where for each $k \geq 1$, $\phi_k(\cdot; \mathcal{V}'^{(i)}) : \mathcal{V} \rightarrow \mathbb{R}$ is an eigenvector of $\Pi_{\mathcal{V}'} \Pi_{\mathcal{V}'^{(i)}} \Pi_{\mathcal{V}'}$, which is a graph signal localized over $\mathcal{V}'^{(i)}$. Let $\Psi(\mathcal{T}'^{(i)}) = (\psi_n(\cdot; \mathcal{T}'^{(i)}))_{n \geq 1}$, where for each $n \geq 1$, $\psi_n(\cdot; \mathcal{T}'^{(i)}) : \mathcal{T} \rightarrow \mathbb{R}$ is a PSWF localized over $\mathcal{T}'^{(i)}$. Each $\psi_n(\cdot; \mathcal{T}'^{(i)})$ is an eigenfunction of $\Pi_{\Omega'} \Pi_{\mathcal{T}'^{(i)}} \Pi_{\Omega'}$. We construct a dictionary candidate as

$$\Xi(t_c, \ell) = \left(\Phi(\mathcal{V}'^{(i)}) \otimes \Psi(\mathcal{T}'^{(i)}) \right)_{i=1}^I, \quad (3)$$

where

$$\Phi(\mathcal{V}'^{(i)}) \otimes \Psi(\mathcal{T}'^{(i)}) = \left(\phi_k(\cdot; \mathcal{V}'^{(i)}) \otimes \psi_n(\cdot; \mathcal{T}'^{(i)}) \right)_{k,n \geq 1}$$

is the set of signals localized over $\mathcal{S}^{(i)}$, $t_c = (t_c^{(i)})_{i \geq 1}$, and $\ell = (\ell^{(i)})_{i \geq 1}$. While dictionaries with analytical forms and predefined parameters capture global signal structure for reconstruction, learned dictionaries with optimized parameters outperform by adapting to observed signals [27]–[29]. We optimize the parameters t_c and ℓ using training samples to learn the dictionary $\Xi(t_c, \ell)$ for signal reconstruction as follows.

Given noisy samples of a generalized graph signal f at a subset $\mathcal{M} \subset \mathcal{V} \times \mathcal{T}$ of vertices and time instances, the goal is to recover the signal f . Suppose the sampling set is $\mathcal{M} = \{(v_m, t_m) : m = 1, \dots, M\}$ and the noisy observations are given by $y_m = f(v_m, t_m) + \epsilon_m$, for $m = 1, \dots, M$, where ϵ_m are independent and identically distributed (i.i.d.) zero-mean noise with variance σ^2 . We assume that the signal of interest is localized in a finite interval $[0, \delta]$ in practice. Denote the training samples $\mathbf{Y}_{\text{train}} = (y_m)_{m \in \{1, \dots, M\}}$, and the dictionary $\Xi(t_c, \ell)$ evaluated at the training samples as $\Xi(\mathcal{M}_{\text{train}}; t_c, \ell)$, where $\mathcal{M}_{\text{train}}$ denotes train instances. The dictionary learning for signal recovery problem is formulated as:

$$\begin{aligned} & \arg \min_{\mathbf{x}, t_c, \ell} \|\mathbf{Y}_{\text{train}} - \Xi(\mathcal{M}_{\text{train}}; t_c, \ell) \mathbf{x}\|_2^2 + \mu \|\mathbf{x}\|_1 \\ \text{s. t. } & t_c^{(i)} \in [0, \delta], 0 \leq \ell^{(i)} \leq \delta, \forall i. \end{aligned} \quad (4)$$

Here, $\mathbf{x}$ contains sparse coefficients, and μ is a regularization parameter. We solve the problem by alternating minimization

over $\mathbf{x}$, $\mathbf{t}_c$ and ℓ until convergence. At the u -th iteration, the following minimizations are performed sequentially:

$$\begin{aligned} \mathbf{x}^{u+1} &\leftarrow \arg \min_{\mathbf{x}^u} \|\mathbf{Y}_{\text{train}} - \Xi(\mathcal{M}_{\text{train}}; \mathbf{t}_c^u, \ell^u) \mathbf{x}^u\|_2^2 + \mu \|\mathbf{x}^u\|_1 \\ \mathbf{t}_c^{u+1} &\leftarrow \arg \min_{\mathbf{t}_c^u} \|\mathbf{Y}_{\text{train}} - \Xi(\mathcal{M}_{\text{train}}; \mathbf{t}_c^u, \ell^u) \mathbf{x}^{u+1}\|_2^2, \\ &\text{s. t. } \mathbf{t}_c^{(i)} \in [0, \delta], \forall i, \\ \ell^{u+1} &\leftarrow \arg \min_{\ell^u} \|\mathbf{Y}_{\text{train}} - \Xi(\mathcal{M}_{\text{train}}; \mathbf{t}_c^{u+1}, \ell^u) \mathbf{x}^{u+1}\|_2^2, \\ &\text{s. t. } 0 \leq \ell^{(i)} \leq \delta, \forall i. \end{aligned}$$

The sparse coefficients $\mathbf{x}$ are updated using Lasso regression [30]. After updating $\mathbf{x}$, the dictionary is updated by optimizing $\mathbf{t}_c$ and ℓ using the gradient descent algorithm [31]. We summarize the complete procedure in Algorithm 1 (details in [26]), referred to as joint energy concentrated dictionary (JECD). Suppose the optimization solution is given by $(\tilde{\mathbf{x}}, \tilde{\mathbf{t}}_c, \tilde{\ell})$. With the optimized dictionary $\Xi(\tilde{\mathbf{t}}_c, \tilde{\ell})$, the reconstruction process is formulated as:

$$\arg \min_{\mathbf{x}} \|\mathbf{Y}_{\text{test}} - \Xi(\mathcal{M}_{\text{test}}; \tilde{\mathbf{t}}_c, \tilde{\ell}) \mathbf{x}\|_2^2 + \mu \|\mathbf{x}\|_1 \quad (5)$$

where $\mathcal{M}_{\text{test}}$ denotes test instances and $\mathbf{Y}_{\text{test}}$ the corresponding test samples. Suppose the optimal solution is given by $\mathbf{x}^*$. The reconstructed generalized graph signal is given by $\hat{f}(v, t) = \Xi(\{(v, t)\}; \tilde{\mathbf{t}}_c, \tilde{\ell}) \mathbf{x}^*$ for each $(v, t) \in \mathcal{V} \times \mathcal{T}$. The recovery performance is measured by the relative square error (RSE) $\frac{\sum_{(v, t) \in \mathcal{M}_{\text{val}}} (f(v, t) - \hat{f}(v, t))^2}{\sum_{(v, t) \in \mathcal{M}_{\text{val}}} f(v, t)^2}$, where $\mathcal{M}_{\text{val}}$ denotes a validation set of sample points sampled distinctly from $\mathcal{M}_{\text{test}}$.

V. EXPERIMENTAL RESULTS

We evaluate JECD on a high-resolution vehicle flow dataset² from 70 stations over two weekdays. Following [32], time series for each highway segment are constructed as direction-adjusted weighted averages of nearby stations.

We randomly select a proportion p_o of Monday's records as training samples to optimize the joint vertex-time dictionary Ξ . Using the optimized dictionary, we reconstruct Tuesday's records from a proportion p_o of Tuesday's samples and evaluate performance on the remaining $1 - p_o$. To achieve maximally localized resolution in vertex domain, we localize the basis functions in the dictionary at each vertex, setting $\mathcal{V}^{(i)}$ in (3) to be $v_i \in \mathcal{V}$ for $i = 1, 2, \dots, N$. We compare JECD against the following joint vertex-time dictionaries for reconstructing generalized graph signals: a) **Joint Fourier transform (JFT)**: A joint vertex-time dictionary is constructed as $\{\phi_k \otimes \psi_l : k = 1, \dots, K, l = 1, \dots, L\}$ [14], where ϕ_k is the k -th orthonormal eigenvector of the graph Laplacian $\mathbf{L}$, and ψ_l is from the orthonormal Fourier basis $\{\frac{1}{\sqrt{2\pi}}, \frac{1}{\sqrt{\pi}} \cos(lt), \frac{1}{\sqrt{\pi}} \sin(lt)\}_{l \geq 1}$ in the time domain. Both K and L are tunable hyperparameters. b) **Short time-vertex Fourier transform (STVFT)** [18]: We design the mother vertex-time kernel in the short time-vertex Fourier basis separably. In the vertex domain, following [18], the localized graph basis is $\mathbf{h}_{p,q} = (\mathbf{U}h(\Lambda - \Lambda_q)\mathbf{U}^T)\delta_p$, where $h(\Lambda) = \sin(0.5\pi \cos((\pi\Lambda)^2))$ is an Itersine kernel [33] and $\Lambda_q = \text{diag}(\frac{\lambda_{\max}}{Q}q)$ for $q = 1, \dots, Q$. Here, $\mathbf{U}$ is

the unitary matrix of Laplacian eigenvectors and δ_p is a Kronecker delta centered at vertex p . In the time domain, we use Gabor functions $g_{m,n}(t) = \frac{1}{\sqrt{2\pi\rho}} e^{-\frac{(t-m\tau_0)^2}{2\rho^2}} e^{jn\omega_0}$ with shifts τ_0 and ω_0 , window width ρ and $m, n \in \mathbb{Z}$. The joint vertex-time dictionary is $\{\mathbf{h}_{p,q} \otimes g_{m,n}(t)\}_{p,q,m,n}$ with tunable parameters Q, τ_0, ω_0 , and σ . c) **Spectral time-vertex wavelet transform (STVWT)** [18]: The separable mother vertex-time kernel in the spectral time-vertex wavelet basis consists of the localized graph wavelet basis $\mathbf{h}_{p,s} = (\mathbf{U}h(s\Lambda)\mathbf{U}^T)\delta_p$, with $h(s\Lambda)$ as a scaled Itersine kernel [33], and the Morlet wavelet $\psi_{a,b}(t) = \frac{1}{\sqrt{a}} e^{-\frac{(t-b)^2}{2a^2}} e^{j\omega_0 \frac{t-b}{a}}$, where $a > 0$ controls dilation, $b \in \mathbb{R}$ shifts time, and ω_0 is the central frequency. The joint vertex-time dictionary is $\{\mathbf{h}_{p,s} \otimes \psi_{a,b}(t)\}_{p,s,a,b}$ with tunable hyperparameters s, a and b .

Fig. 2a shows the reconstruction performance across training ratios p_o from 0.03 to 0.15. JECD consistently achieves the lowest RSE, demonstrating superior reconstruction accuracy. At lower training ratios (0.03 to 0.07), it shows a sharper RSE decline, indicating better learning efficiency from limited data. As the training ratio increases, JECD maintains a low and stable RSE, highlighting its robustness. Fig. 2b further demonstrates this by evaluating performance across different Signal-to-Noise Ratio (SNR) levels with a fixed training ratio of 0.1. JFT performs worst across all SNR levels, while STVFT and STVWT perform better than JFT but lag behind JECD, particularly at lower SNRs. These results emphasize JECD's effectiveness in handling noise with limited training data.

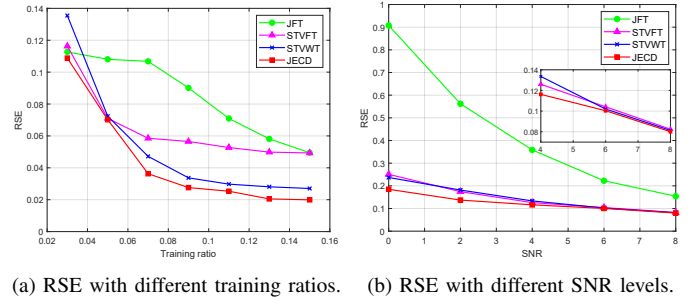


Fig. 2. Reconstruction error under varying conditions. Each point represents the average of 10 repetitions.

VI. CONCLUSION

We have derived an uncertainty principle for generalized graph signals, unifying classical time-frequency and graph uncertainty principles. We have introduced the joint vertex-time and spectral-frequency spreads to measure signal localization and identified a class of signals with maximal energy concentration in the vertex-time domain and localization in the spectral-frequency domain. These form a new joint vertex-time dictionary. This dictionary demonstrates superior reconstruction accuracy and noise robustness in real-world experiments, compared to existing baselines. Future work will focus on developing a time-varying dictionary to handle more complex and dynamic scenarios, optimizing its adaptability and performance in non-stationary environments.

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